August 24, 2026

Dear StackAdapt Hiring Team,

I am writing to apply for the Machine Learning Engineer position at StackAdapt. With a Master of Science in Computer Science from the University of Calgary and hands-on experience building Python/C++ systems for machine learning, large-scale data processing, performance-sensitive research software, and backend APIs, I am excited by the chance to contribute to a remote-first team working on real-time ML systems for a high-scale advertising platform.

What draws me to StackAdapt is the combination of applied machine learning, low-latency engineering, and data systems. My graduate research was not only about training models, but also about building the surrounding software needed to process, structure, retrieve, and evaluate large datasets reliably. As a Research Software Developer with BigGeo/Mitacs, I built C++ and Python pipelines for Earth observation data, including DGGS-based workflows that converted satellite imagery into structured tensor data for multiresolution storage and retrieval. I also improved one C++ processing workflow from 233 seconds to 26 seconds for a 10-tensor workload through multithreading and performance-focused implementation.

I also bring practical ML experience from my DEM super-resolution work, where I trained PyTorch ResNet models, built data collection and preprocessing workflows using Google Earth Engine and Rasterio, and evaluated model quality with terrain-aware metrics such as RMSE, slope, TRI, TPI, and wavelet-based errors. That project strengthened my understanding of the full model workflow: data quality, sampling strategy, training, evaluation, visualization, and careful interpretation of model behavior.

My software background also includes backend engineering. As a back-end developer, I built Django REST APIs for NLP, text-to-speech, and speech-to-text services backed by PostgreSQL, implemented authentication flows, and contributed to a WebRTC-based chatbot system using GPT-style responses and FAQ data. As a teaching assistant, I supported students in big data applications, AWS-based data pipelines, data-at-scale workflows, SQL, and database systems, which helped me communicate clearly while debugging real implementation issues.

I have not yet worked in production adtech or on ML systems serving thousands of models concurrently, but the core problems in this role feel very close to the systems I want to keep growing into: custom ML algorithms, scalable data pipelines, reliable model workflows, latency-aware engineering, and collaboration between data science and software engineering. I bring strong Python and C++ foundations, PyTorch experience, statistical evaluation habits, backend/API experience, and a serious interest in building ML systems that are not only accurate, but practical and maintainable at scale.

Thank you for considering my application. I would welcome the opportunity to discuss how my background in ML workflows, scalable data processing, performance optimization, backend development, and research software can contribute to StackAdapt's Data Science team.

Sincerely,
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